

Exercises

1. Why is f not a function from $\mathbb{R}$ to $\mathbb{R}$ if
a) $f(x) = 1/x$ b) $f(x) = \sqrt{x}$ c) $f(x) = \pm\sqrt{x^2+1}$?

Solution: a) $0 \in \mathbb{R}$. $1/0$ is undefined. $f(x) = 1/x$ is not defined at $x=0$.

b) The square roots of negative reals must exclusively be complex, not a real number. $\therefore$ If $x < 0$, $f: \mathbb{R} \rightarrow \mathbb{R}$ is not a function s.t. $f(x) = \sqrt{x}$.

c) $f(x)$ has two values for each x , $+\sqrt{x^2+1}$ and $-\sqrt{x^2+1}$. This violates the definition of a function where each ^{element} ~~value~~ in the domain is mapped to a unique element in the codomain.